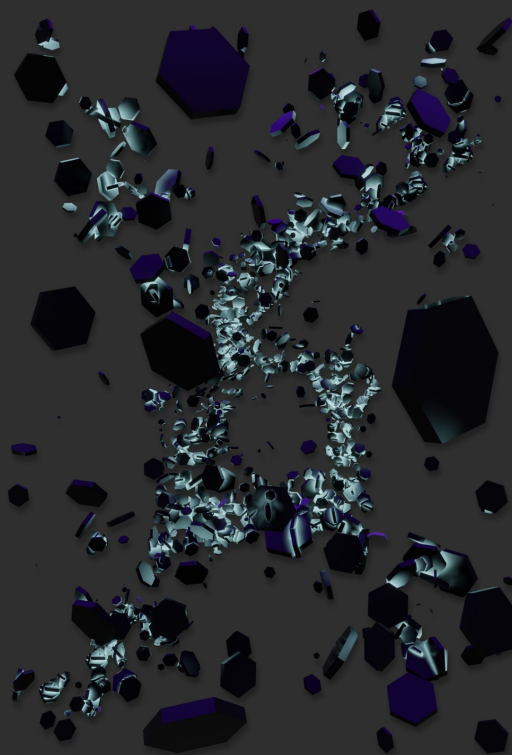




Dissecting Complex Human Behaviour: From the lab, through the clinic and beyond

T. Quendera



Dissertation presented to obtain the **PhD degree in Neuro-
science**

INDP - Champalimaud Research

Oeiras, 2025

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INDP - Champalimaud Research — Instituto de Tecnologia
Química e Biológica — Universidade Nova de Lisboa

Research work coordinated by:



Oeiras, 2025

Author Contributions and Financial Support

Acknowledgements

Abstract

an abstract

Thesis Overview

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Glossary

1 General Introduction

1.1 Introduction

Human behaviour has long been one of the most studied phenomena in nature. Even before the dawn of psychophysics in the 1860s [Fechner, 1948], there have been countless studies and even fields of research attempting to make sense of what and how we do things. These efforts are often focused on specific tasks: on the one extreme, tasks developed to study specific behaviours (i.e. a forced choice Decision Making task), and on the other, more naturalistic tasks like learning how to walk or read. How can we make sense of behaviour is way too broad of a question to focus on. Instead, in this work, I will focus on a battery of tasks - from the very simple to the complex, from learning to learned, from the lab to the clinic - and attempt to shine some light on what human behaviour can look like in them.

1.2 Decision Making and Psychophysics

One of the prominent fields of research of Behavioral Sciences is that of Decision Making. Here, scientists adventure to model, create and test environments that operate under specific constraints. These task constraints constitute the tools with which we can ask questions about specific behaviours. As an example, a two alternative forced-choice task may ask participants to tell wether a a cloud of moving dots is moving rightwards or leftwards [Newsome et al., 1989]. As one approximates a dot motion coherence of 50%, so should the accuracy of responses diminish. In this very simple case, by looking at the psychometric curve of response accuracy given the stimulus orientation, one can

begin to answer the fundamental question of psychophysics: what is the relationship between stimuli and perception.

1.2.1 How do we make decisions

After defining our task, the readout of a decision is most often behavioural: a movement, a verbal answer, the press of a button. What leads to these behaviours is one of the most prolific areas of the study of Decision Making.

1.2.1.1 Neural basis of Decision Making A multitude of brain regions are involved in Decision Making. Depending on the task, certain sensory organs and corresponding cortical areas are recruited. Furthermore, the Anterior Cingulate Cortex (ACC), Orbitofrontal Cortex (OFC) and the Ventromedial Pre-frontal Cortex (VmPFC) are all regions known to be involved in Decision Making. The OFC appears crucial for hidden state representation, and hence for inference-based decisions [Rolls, 2004] [Vertechi et al., 2020]. For example, in both primates and rats, the inhibition or lesion of the OFC seems to impair performance in reversal learning tasks [Wilson et al., 2014]. The ACC, on the other hand, is thought to be involved in integrating value [Williams et al., 2004]. The VmPFC seems critical for the representation of both reward and value [Hiser and Koenigs, 2018]. Most studies thus tend to focus on these (and adjacent) regions as they attempt to highlight the underlying circuitry in Decision Making.

1.2.1.2 Modeling the problem To make sense of both behavioural and neural findings, one can opt to normatively formalize the problem (i.e. model them). Modeling a Decision Making task requires

a deep understanding of its environment. Often, when modeling a task, it's crucial to consider what an optimal agent would behave like.

In realistic scenarios, subject behaviour is often sub-optimal. An ideal model is such that, by manipulating its parameters, one is able to replicate both optimal and sub-optimal (i.e. subjects') behaviour. Once again taking our random moving dots task as an example, one can model an optimal agent with a simple sigmoid/logistic function and account for human error by tuning the steepness of its curve. While such elegant models are a rare sight to behold, the idea of conceptualizing a problem mathematically is one of the most powerful ways one has to make predictions about behaviour and an essential component in most contemporary scientific studies.

1.2.2 How do we learn to make decisions

The decision-making process often starts with sensory input from the environment, such as visual, auditory, or tactile information. Sensory receptors collect this data and transmit it to the brain. [Mesulam, 1998] The brain's sensory cortices (e.g., the visual cortex for visual information) process this raw data into a coherent perceptual experience. This involves integration and interpretation of sensory input to form a meaningful context. Upon reaching the cortex, the information is evaluated by the prefrontal areas (see 1.2.1.1).

Recruiting the Basal Ganglia and the Dopaminergic system - where action selection and reward prediction errors are computed respectively - our brains then use Reinforcement Learning (RL) mechanisms to adapt and refine Decision Making over time [Schultz et al., 1997].

Then, memory comes into play. Areas like the hippocam-

pus and the entorhinal cortex contribute to memory consolidation [Squire and Zola-Morgan, 1991].

This description is equal parts incomplete as it is reductionist.

1.2.2.1 We don't all learn the same However known these neural systems are, there is always variance in observed behaviour. Learning differences may be due to different factors: Genetic, Neurobiological, Emotional, Neuroplasticity, etc. [Zatorre et al., 2012] [Ericsson et al., 1993]. Learning differences may be one of the explanatory causes for the behavioural variability that is very often observed on any particular task. Recent discoveries like the complete neural connectome for several animal species, have highlighted what we suspected long ago: there are many a paths through which information flows in the brain [Asahina et al., 2022]. Conversely, it is increasingly accepted that even the simplest of neural circuits can lead to a variety of behavioural outputs [Cruz et al., 2022]. Simultaneously, environmental changes in behaviour can significantly alter behaviour. As an example, zebrafish alter their turning behaviour based on the temperature of the water in which they grew in [Hageter et al., 2021]. Social interactions and hierarchies can also change behaviour, as it has been demonstrated across different mammalian species (e.g. in rodents [Faure et al., 2022]).

Ultimately - and given your specific genome, neural circuitry, development circumstances, social interactions, etc. - it is by analyzing the current state of the environment and taking into account its past states that one learns and decides on how to solve a task.

1.2.2.2 Inter and Intra-Subject variability While it may seem like a trivial affair, distinguishing inter and intra-individual variability requires a large amount of data - and repeated measures - from a given individual. What are individual behavioral differences and what are differences across individuals is a question that is relevant in any study. Big Data approaches have recently been found useful to address this type of problem.

1.3 Complexity

As we distance ourselves from discrimination tasks, both the underlying neural mechanisms needed to produce an answer and plausible models of the data become ever more complex. While a simple sigmoid suffices to model behaviour in the random moving dots task and information from neurons in the visual cortex alone is enough to solve it [Newsome et al., 1989], this is not the case for the large majority of tasks. Coming up with tasks that are simple to model and solve and yet complex enough to elicit interesting responses is thus one of the greater challenges in the study of Decision Making.

1.3.1 What does it mean to be complex

One definition of complexity is how large the dimensionality of a problems' state-space is. Van Opheusden and Ma (2019), have attempted to quantify a battery of problems in terms of their complexity. In their study, they confirmed something that has long been hinted at: Neuroscience tends to stick to low-complexity problems. In contrast, the field of Artificial Intelligence (AI) - full to the brim with data - tends to focus on extremely complex tasks. This begs the question as

to what lives in the middle ground, and why are we not exploring it.

1.3.1.1 The curse of dimensionality With increasingly complex questions, so does the amount of data required to answer them increase. This, in other words, is the problem that Bellman [Bellman et al., 1957] highlighted and named *the curse of dimensionality*. In it, it is stated that, as the dimensionality of a problem increase, data points become ever more sparse, making it difficult to discern meaningful relationships. To thoroughly sample a larger space, exponential amounts of data are required. This is an especially relevant problem in the fields of AI and Machine Learning (ML).

This curse is not without its cure. Reducing the dimensionality of the dataset is the solution scientists have found to deal with the problem. Be-it by manually feature extracting important factors before analysis or by utilizing dimensionality reduction techniques (e.g. Principal Component Analysis (PCA)).

1.3.1.2 Discerning a signal amongst noise This is however easier said than done. Manually extracting the relevant features requires the analyst to make assumptions about the dataset. Determining what is signal and what is noise is an extremely important decision. Dimensionality reduction techniques aren't necessarily an instantaneous solution either: 1) the linear combinations of the original dataset often lead to a loss of data; and 2) it becomes difficult to determine which features are the most significant in the dataset.

1.3.2 Why studying complexity matters

Handling complex datasets seems to be easier said than done, so why is there even an attempt to do so?

Firstly, some problems are inherently complex. In fact, most naturalistic behaviours are. One of the most explored fields is that of language. While heavily theorized on, it was in fact in the last decades that language studies datasets have had a resurgence. Statistical language modelling - or the capability to generate language from observed data - saw it's first steps in the 1980's. Quickly, scientists ran into the curse of dimensionality. However, and synchronously, a new technology grew - the Internet. Developers from Google were quick to understand that this "dataset" was unrivaled and that leveraging it was the way forward. Indeed, in 2018, they launched BERT - a flagship model that learns to represent text as a sequence of vectors probabilistically similar to the previous. Today, with the rise of the GPT-family models, Large Language Models (LLMs) have quickly become ubiquitous. These models rely on dozens of billion parameters and leverage the latest advances in neural network algorithms to generate text in natural language. Natural Language Processing has dramatically been improved and its methods radically altered by the availability of this corpus of data and algorithms [Guo et al., 2025].

Other notable examples of success due to large datasets are that of Stockfish and Dall-e (an algorithm that solves the game of chess and an image generation algorithm respectively). These are, however, not the standard for psychology .

1.3.2.1 Learning from Psychophysics As highlighted before, in psychophysics, simple tasks are boiled down to an (or several) essential measurements that succinctly describe behaviour. This powerful approach allows for comparisons in and across subjects.

Suffice to say, this methodology does not come without some qualms:

1. It is often hard to reduce a behavioural phenomena that may be extremely complex to a single measurement.
2. Psychophysics tasks often require large repetitions of trials in cumbersome settings, which can lead to fatigue.
3. Sensory adaptation can occur through exposition to a task, making measurements unreliable.
4. Individuals may behave in drastically different ways, impeding generalizable measures.
5. Individuals may have different perceptions even when present with the same stimulus, which can lead to radically different answers.

All of these must be taken into account when investigating behaviour and applying these tools. Our ambition is to be rigorous and extend these tools to increasingly complex datasets without falling into its pitfalls.

1.3.2.2 Extending from Psychophysics One way to extend these tools to account for more complex tasks is to start simple. In a game of chess, one might be compelled to understand how to win. Instead, we propose starting with the first move. This approach comes with several advantages.

1. It maximizes data: all games start on the first move.

2. It ensures comparability across subjects.
3. It ensures the root of the behavioral tree is explored before dwelling into the branches: no assumptions are made about the more complex aspects of the data.

Following this approach, we can slowly navigate the layers of complexity of any given task.

1.3.2.3 Do we have the tools and the means? An approach like the proposed may seem unrealistic, depending on the task. A critic might even say that learning about the first move would lead nowhere, if the goal is to understand chess. The single most crucial point thus becomes choosing our research question. We must be aware of the limitations of our approach and design/choose tasks that thrive in them.

1.3.2.4 Modelling complexity One of the tools at our disposal is to compare our investigations with models. Starting with the simplest possible - 0th order model - and growing our models in complexity as needed. In a game of chess, we can imagine that the simplest possible model for the first move is to play a random piece. It should be immediately apparent why this approach is not as simple as it seems. All tasks have rules, and accounting for them is needed. One cannot simply move any chess pieces (e.g. bishops don't go through pawns.) Building the environment first is the needed missing step. We must create an environment to test our models on that follows exactly the rules of our task. That means our task must be computationally formalized, and while most things are, usually it does not come without its challenges (e.g. the rules of chess alone require between two to

three thousand lines of code.)

1.3.2.5 “AI”

1.3.3 The clinic as a fertile ground for studying variability

1.3.4 Curse of dimensionality

2 Behaviour in the lab

2.1 Introduction

2.1.1 Decision Making

Foraging is a fundamental behavior observed across species, involving complex decision-making processes that balance the exploitation of known resources with the exploration of new opportunities. Animals, including humans, must continually assess whether to persist in a given location or shift to another in search of better rewards. This trade-off, often framed as the exploration-exploitation dilemma, has been extensively studied in behavioral ecology, neuroscience, and psychology [Cohen et al., 2007, Hills et al., 2015].

2.1.1.1 Inference Based vs. Stimulus Bound Behaviour

While some decision-making models propose that foraging behavior is driven by direct reinforcement from past rewards, others suggest that individuals use inference-based strategies to estimate the underlying availability of resources [Behrens et al., 2007].

Recent studies on rodent foraging behavior suggest that animals infer hidden states of resource availability rather than merely responding to reinforcement contingencies [Vertechi et al., 2020]. However, it

remains unclear whether humans engage in similar inference-based decision-making when faced with analogous foraging challenges. To investigate this, we adapted a rodent foraging task to a human experimental framework, allowing for cross-species comparisons and a deeper understanding of the cognitive mechanisms underlying foraging decisions.

To systematically explore these questions, we conducted two experiments. The first experiment examined how changes in task difficulty influenced foraging behavior by varying reward and transition probabilities. The second experiment investigated the stability of foraging strategies over time by inviting participants for multiple sessions. This allowed us to assess whether human decision-making in this task is consistent across time, an essential consideration for understanding the reliability of foraging strategies [Daw et al., 2006].

Our findings contribute to cross-species decision-making by elucidating how humans integrate uncertainty, past experiences, and task constraints when making foraging decisions. Understanding these mechanisms has implications for fields ranging from behavioral economics to artificial intelligence, where optimal decision-making under uncertainty is a crucial challenge [Gershman, 2018, Sutton et al., 1998].

The remainder of this chapter details our experimental design, behavioral results, and a discussion of the implications of our findings for theories of decision-making and foraging behavior.

2.1.2 Designing experiments across species

The integration of human and rodent research in neuroscience is crucial for understanding fundamental brain mechanisms and translating

findings into clinical applications. Rodents, particularly mice and rats, provide a controlled experimental framework for investigating neural circuits, genetics, and behavior at a level of detail not feasible in human studies [Gomez-Marín and Ghazanfar, 2019]. Their well-mapped neuroanatomy and genetic tractability allows researchers to manipulate specific neural pathways and assess causal relationships between brain activity and behavior. However, translating rodent findings to humans requires careful validation, as behavioral and cognitive processes can differ across species [Charvet et al., 2025]. Comparative approaches, such as cross-species tasks assessing decision-making or learning, help bridge this gap by identifying conserved mechanisms while accounting for species-specific adaptations [Vertechi et al., 2020]. This synergy between rodent and human research is particularly valuable in areas like neuropsychiatric disorders, where preclinical models inform treatment strategies, but human validation remains essential for clinical relevance. By leveraging both human cognitive complexity and rodent experimental precision, neuroscience can more effectively unravel the principles governing brain function and behavior [Krakauer et al., 2017].

2.2 Materials and Methods

2.2.1 The "Hidden-State" task

In natural foraging behaviors, animals are faced with a constant dilemma: whether to stay and exploit resources at their present location or , with associated costs, to venture out and explore alternative, potentially more beneficial sites. When viewed through the lens of optimal decision-making, the key question is to determine the most advantageous moment to abandon the current site in favor of the next. To

gather insight on this process, a simple behavioral task was developed in mice [Vertechi et al., 2020]. In it, at any given moment, rewards are available at one of two distant foraging sites. For each exploration on the rewarded site, rewards are administered following a probability $P(\text{reward})$. Simultaneously, for each exploration, rewards change site availability following a probability of transition $P(\text{transition})$. Crucially, these are not explicitly available to the agent - they must be inferred. Rewarded trials occur when foraging on an active site where $P(\text{reward})$ was triggered. Unrewarded trials, however, aren't as clear. One must estimate, for any given trial - taking into account the task history - whether an exploration was unrewarded because rewards are currently available only on another site or whether an exploration was just "unlucky" (i.e. did not trigger $P(\text{reward})$). Subjects are thus tasked with inferring a hidden state - reward availability - from a sequence of rewards and failures.

2.2.1.1 Adapting the task to humans To allow for behavioral comparison across-species, the task was adapted to humans. We developed a video-game like task where subjects were instructed to control a character on a touch-screen device. This character was tasked with finding and hitting a "monster". This "hidden-state monster" hid in one of two walls (foraging sites), separated by a costly to travel on path. Hitting the monster is analogous to getting a reward - awarded in game points. The cost of traversal was manipulated by making the path slower or faster to travel across. $P(\text{reward})$ and $P(\text{transition})$ were kept equivalent to the mouse counterpart.

2.2.1.2 How to solve it There are several ways in which to integrate rewards and failures towards a decision. In a stimulus-bound process, the value of a site is directly related to the rewards on that site, decreases with rewards on the other side and decays to 0 with failures. An alternative process is that of using inference to solve the task: an agent that is aware of the fact that a hidden state determines reward availability can use rewards and failures differently to determine whether the current site is ripe or not. Both stimulus bound and inference models predict that the probability of leaving a site increases with failures, however, the way rewards are represented is quite different [Vertechi et al., 2020]. In a stimulus bound process, accruing rewards on a given side would increase the value of said side and thus increase the likelihood of remaining. In an inference based process, rewards reset the probability of leaving a site to 0, as it provides the agent with certainty about the hidden state.

2.2.2 Describing the experiments

Human participants were invited to participate in this research study. Their participation varied across several experiments.

2.2.2.1 Human vs. Mouse In the data used to compare directly to mice, participants were invited for a single visit to the lab. In this work, we will use this dataset independently of the rodent data to look at how changes in the task modulate human behaviour. For a comprehensive comparative study refer to [Vertechi et al., 2020].

After informed consent, participants were presented with the experimental protocol. In it, they experienced the task a total of 4

times. $P(\text{reward})$ and $P(\text{transition})$ were manipulated in the following protocols:

	Probability of reward	Probability of transition
Medium	90	30
Easy	90	90
Hard	30	30

Table 1: Protocols and respective settings

All the visits started with the medium protocol and randomly interleaved the easy and hard protocols (e.g. Medium - Easy - Medium - Hard).

2.2.2.2 Humans across time In another experiment, human subjects were once again invited to the lab. In this experiment, participants were invited to two separate visits in time. The goal was to assess whether the task findings were consistent across time. For this experiment, and following our initial findings of how fast humans learn the task, a protocol of $P(\text{reward}) = 45$ and $P(\text{transition}) = 15$ was adopted. This protocol is not necessarily harder, as it retains the same ratio of $P(\text{reward})$ and $P(\text{transition})$. Nonetheless, it should require more attempts after last reward. Participants interleaved the medium protocol with the new protocol.

	Probability of reward	Probability of transition
Medium	90	30
New	45	15

Table 2: Medium and New protocol settings

2.3 Results

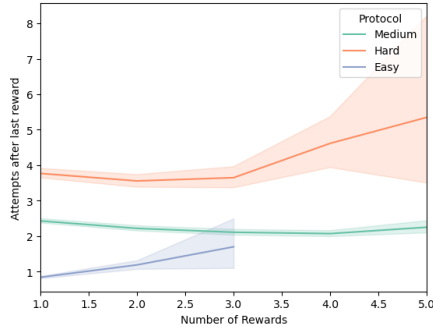
2.3.1 Behaviour is maintained across changes in the task

As demonstrated in [Vertechi et al., 2020], changes in protocol were reflected in consistent behavioural changes.

This task investigates if participants solve it in an inference-based or stimulus bound fashion. To investigate whether that was the case, we look at reward history. In a stimulus bound process, accruing rewards on a given side would increase the value of said side and thus increase the likelihood of remaining. In an inference based process, rewards reset the probability of leaving a site to 0, as it provides the agent with certainty about the hidden state. In an inference based process, the number of attempts on a given side after the last reward was administered should thus not be reliant on the reward history. In Figure 1 we demonstrate that, independently of protocol, the total number of rewards has no effect on the amount of attempts after the last reward.

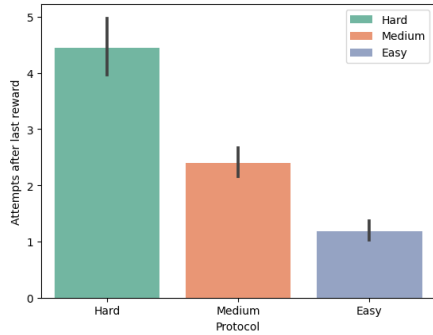
Next, we investigated whether this task was sensitive to changes in protocol. For all three protocols, in figure 2, we show the mean attempts after the last reward was administered and respective 95% confidence intervals error. What is immediately salient is that the number of attempts after the last reward scales with the difficulty of the task - the harder the protocol, the more evidence do our participants procure. That is congruent with our hypothesis that human subjects use an inference-based process to solve this task.

In figure 3, you can observe the adaptation of players to the protocols. As subjects always start with the 'Medium' protocol, it's value



The mean attempts after last reward (line) and 95% confidence interval (shaded), per protocol, given the number of total rewards acquired.

Figure 1: Attempts after last reward are independent from rewards

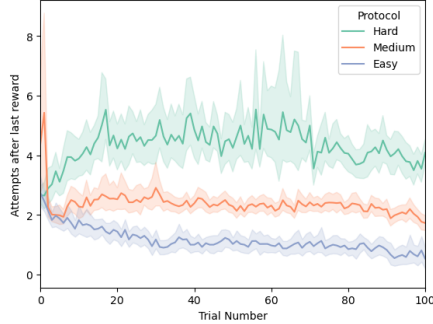


In colors, the mean attempts after last reward (in order, hard, medium and easy protocols). In a black line, the 95% confidence intervals for each respective protocol.

Figure 2: Attempts after last reward for all protocols

of attempts after last reward is unstable for the first trials, yet, coalesces to a stable value quickly. In contrast, because both the 'Easy' and 'Hard' protocols succeed the 'Medium' one, their value of mean attempts after last reward begin where the 'Medium' protocol is stable.

These are hints that not only our subjects adapt to different protocols but also that their adaptations make sense given the experimental design. Figure 3 also hints that subjects learn fast - a feature highlighted in the original manuscript as exclusively human [Vertechi et al., 2020].



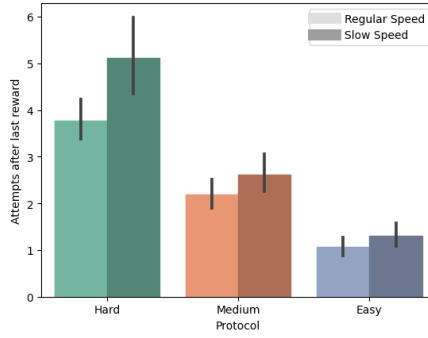
The attempts after last reward as trial number increases, divided by protocol.

Figure 3: Attempts after last reward over trials

One other manipulation of task parameters was done on the time to travel between reward locations. To do so, the speed of traversing between locations was diminished in some trials, interleaved with regular speed trials. As illustrated on figure 4, we found that decreasing the speed of traversal led to an increase in the number of attempts after the last reward.

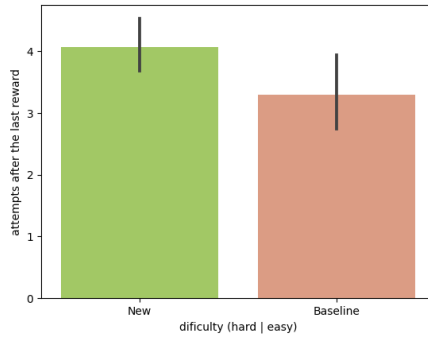
2.3.2 Behaviour is maintained across sessions

In the second experiment, subjects were invited to two visits. In each of the visits they were presented with two protocols - "medium" and "new" - in a randomized order. Because the ratio between the probability of rewards and of site availability is maintained, the protocol should be



The attempts after last reward per protocol with or without a speed impairment (shaded).

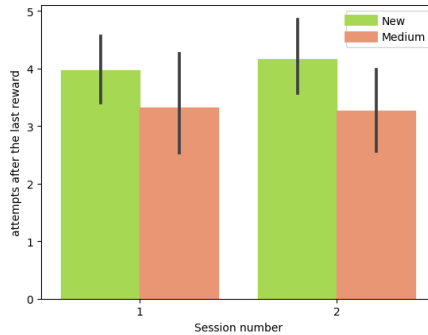
Figure 4: Traversing speed impacts the attempts after last reward of similar difficulty. However, due to less rewards being available with lower probabilities - in the novel scenario -, the amount of attempts after last reward increases. In figure 5, we plot the average attempts after last reward for both protocols for both sessions combined.



The attempts after last reward per protocol and independently of session. Compared to the baseline protocol, the new protocol produces more attempts after last reward. Black line signifies the 95% confidence interval.

Figure 5: The new protocol produces more attempts after last reward

The novelty of this experiment was the introduction of a follow up session. In Figure 6 we see the average attempts after last reward for both protocols and both sessions. Confirmed by Figure 7, the mean attempts after the last reward was maintained across both sessions in the same individuals. An Intra-Class Correlation analysis (table 3) demonstrated a good reliability of the measure across the two visits. In the bland-altman plot in Figure 8, the average difference in attempts after the last reward across the two visits is shown. In it, we can visualize whether there is systematic bias (consistent difference between the two visits) and whether the agreement between the two visits changes across the range of measurements. The measurement stays largely inside bounds. Other than that, some spread can also be observed, indicative of heteroscedasticity: A trend in the data seems to indicate that with larger average attempts after last reward values across both visits, the measure reliability decreases.



The average difference in attempts after last reward across the two visits.

Figure 6: The average difference in attempts after last reward across the two visits

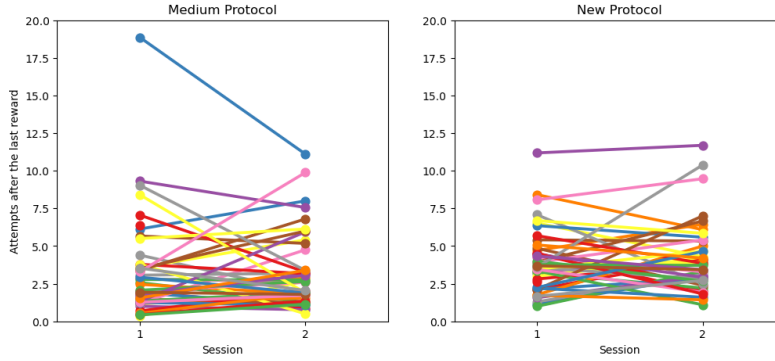


Figure 7: The average difference in attempts after last reward across the two visits

On the left panel, the medium protocol and on the rightmost, the new protocol. Colors are different participants, columns are visits.

ICC	p-value
0.833536	1.219174e-08

Table 3: Intraclass correlation analysis

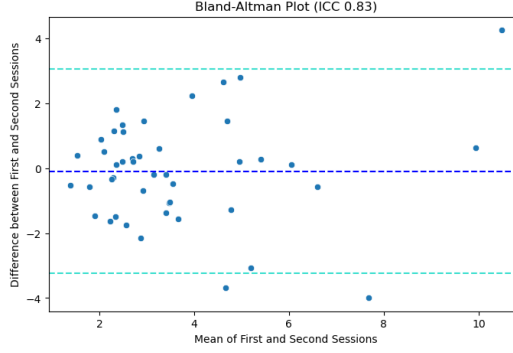


Figure 8: The average difference in attempts after last reward across the two visits.

Bland-altman plot for the attempts after last reward across the two visits. A horizontal line is drawn at the mean of the differences. This line represents the average bias between the two visits. Two additional horizontal lines are drawn at mean difference ± 1.96 standard deviations. These are the limits of agreement and represent the range within which 95% of the differences between the two methods are expected to lie. A trend of heteroscedasticity can be observed - the differences between the visits becomes larger as the magnitude of the attempts after last reward increases.

2.4 Discussion

This first chapter aimed to investigate whether human participants solve a foraging task using an inference-based or stimulus-bound decision-making process. By adapting a behavioral paradigm originally designed for mice, we examined how human participants responded to changes in reward probability, transition probability, and traversal costs. Our results suggest that participants employ an inference-based strategy, adjusting their behavior according to the task demands and maintaining stable performance across different protocols and experimental sessions.

2.4.1 Inference-based decision-making in human foraging

The findings align with previous research [Vertechi et al., 2020] demonstrating that humans use inference-based strategies to estimate reward availability. This is evident in the fact that, regardless of the total number of rewards obtained, participants did not increase their persistence on a given site. Instead, their probability of leaving a site was reset after receiving a reward, supporting the notion that rewards serve as confirmatory evidence of the hidden state rather than directly reinforcing a given location. This contrasts with a stimulus-bound strategy, where the value of a location would increase with the accumulation of rewards, leading to site persistence.

As the difficulty of the protocol increased, the number of attempts after the last reward also increased. This further supports the inference-based decision-making framework. Participants required more evidence before switching sites when the task was more uncertain, which suggests they were actively inferring the hidden state rather than simply reacting

to immediate rewards and failures in a stimulus-bound fashion.

2.4.2 Task sensitivity and behavioral adaptation

The results indicate that human participants adapt flexibly to task changes. The different protocols elicited systematic variations in behavior, with participants making more attempts after the last reward in harder protocols. This aligns with the hypothesis that participants adjust their exploration-exploitation trade-off based on task demands. Moreover, the learning curves suggest that participants rapidly adapted to new conditions, particularly as they transitioned from the medium protocol to either the easy or hard protocols. This highlights the flexibility of human decision-making and the ability to integrate prior experience into ongoing behavior.

The effect of traversal cost on decision-making further supports this notion. When movement between foraging sites was slowed down, participants increased their persistence at a given location before switching. This finding suggests that participants integrate costs into their decision-making process, optimizing their strategy based on both reward probabilities and movement constraints. Such behavior is consistent with an optimal foraging framework, in which organisms seek to maximize reward while minimizing costs.

2.4.3 Stability of behavior across sessions

The second experiment clarifies that participants' decision-making strategies remain stable over time. The number of attempts after the last reward remained consistent across sessions, and the intraclass correlation analysis confirmed strong reliability of this measure. The

Bland-Altman analysis further suggested that individual differences in behavior were maintained across visits, with minimal systematic bias. However, some heteroscedasticity was observed, indicating that reliability decreased slightly for individuals who had higher values of attempts after the last reward.

This consistency over time underscores the robustness of the inference-based approach and suggests that individuals develop and retain an internal model of task dynamics that guides their decisions across sessions.

2.4.4 Conclusion

Our findings have several important implications for the study of decision-making.

The consistency of behavior across different conditions suggests that these strategies are flexible and context-dependent. Understanding how humans balance exploration and exploitation has implications for fields such as behavioral economics, neuroscience, and artificial intelligence, particularly in the development of models that seek to mimic human decision-making.

The demonstrated behavioral patterns indicate that decisions are based on inference rather than simple reinforcement learning. Participants flexibly adapted their behavior to task demands and maintained stable decision-making strategies across multiple sessions. These findings contribute to our understanding of human exploration-exploitation trade-offs and offer a framework for future studies on the cognitive mechanisms of decision-making in uncertain environments like the one shown in the following chapter.

2.5 Author Contributions

- Rodent task design by P. Vertechi, D. Sarra and E. Lottem.
- Human adaptation of the task by T. Quendera and P. Vertechi.
- Running the experiment by T. Quendera and C. Seybert.
- Coding of the task, analysis and discussion by T. Quendera.

2.6 Acknowledgements

I would like to acknowledge the incredible work done by my colleagues in the rodent team that enabled me to produce such clean results. A special shout-out to Pietro Vertechi for being the best guide and tutor that I could have hoped for when adapting the task to humans. Without the patience and guidance of all the rodent team this translational work would not have been possible. Thank you to the Mainen Lab, especially Z. Mainen in supporting this research and guiding me through it.

A special thank you to the Neuropsychiatry unit team for running the experiment when I could not for the longitudinal phase of this project. The work hours they put into supporting this project are invaluable and the major reason why it even exists. Thank you to C. Seybert, G. Cotovio and A. Oliveira-Maia and all the NPU for the guidance and help throughout this project.

3 Behaviour in the clinic

3.1 Introduction

3.1.1 Why translational research?

3.1.1.1 Obsessive Compulsive Disorder and task performance

3.2 Materials and Methods

3.2.1 The experiment

3.2.1.1 The subjects

3.2.1.2 Obsessive Compulsive Disorder Our clinical cohort consisted of patients diagnosed with Obsessive Compulsive Disorder (OCD). These patients were recruited as a part of a larger comprehensive study on OCD and markers of diagnosis - Neurocomp. This study was approved by the FChampalimaud ethics committee and conducted in the Neuro Psychiatry Unit under the supervision of Dr. Albino Oliveira-Maia. Patients were referred to our research team by their physicians and controls were recruited to match them in demographics. In this study, participants underwent several experiments: one of which, a battery of behavioural tasks on which we will focus on in this work.

3.2.2 The tasks

For this protocol, three different tasks were selected. Two for historical correlations with OCD and one for it's novelty.

3.2.2.1 Reversal Learning Task Firstly, a classical reversal learning was adapted from the work of [Remijnse et al., 2006]. Here, a deficit in reversal learning and aberrant OFC-striatal and dorsal pre-frontal activity was correlated with the diagnosis of OCD.

In each trial, the task consisted of 2 stimuli images presented at either side of a screen. The images had randomized locations and were shown for a maximum time of 3000 milliseconds. Participants selected one of the images by pressing the right or left keys on a keyboard. On a correct response, rewards - an increase in points - were administered 80 percent of the time. For the remainder 20 or for spontaneous errors (SE), a punishment - decrease in points - was administered. A correct response with a rewarded outcome was named a Correct Response (CR). A response that albeit correct was probabilistically attributed a punishment was named a Probabilistic Error (PE). PE's could lead to either a shift or a non-shift in the choice: Probabilistic Error Shift (PES) or Probabilistic Error No Shift (PENS).

Criterion for the trigger of a reversal happened stochastically after 6 to 10 consecutive correct responses. After reversal, a response that persisted on the pre-reversal stimulus was deemed a Preceding Reversal Error (PRE). The last response before an error was named a Final Reversal Error (FRE).

Trials ended with a 2000 millisecond feedback window where the outcome was displayed, followed by a fixation cross of 1000ms and a subsequent trial.

Subjects were instructed to accrue as many points as possible. Speed was not encouraged.

A neutral baseline trial (BL) was also presented randomly inter-

leaved with the probing trials. This trial consisted of two images of a ball and a pencil. Here subjects were instructed to always select the option "pencil". In these trials, a neutral feedback was given.

Subjects were excluded if they did not meet a certain performance level (a minimum of 5000 points), or if they, on average, answered wrongly on BL trials. A total of 15 subjects (8 healthy and 7 ocd) did not meet the criteria.

3.2.2.2 Stop-Signal Reaction Time Task Secondly, a stop-signal reaction time task (SSRTT) was adapted from [Aron et al., 2003]. In this family of tasks, Participants are asked to make speedy answers on go trials and withhold from responses on no-go trials. In the SSRTT, most trials are go trials. However, in a fraction of trials, a tone will be played. When the tone is played, participants should withhold their response.

In this version of the task, an arrow to the left or to the right was presented. The order in which right- and left-pointing arrows were presented was randomized to ensure an unbiased distribution. Subjects were instructed to press the correspondent key on a physical keyboard. In 25% of the trials, chosen at random, the audible stop signal was played shortly after the arrow was shown. The interval between the initial presentation of the arrow and the subsequent stop signal tone is known as the stop-signal delay (SSD).

To keep the task challenging, the inter-stimulus interval (the time between successive stimuli) and the SSD were dynamically adjusted based on each participant's performance. This adjustment aimed to ensure that participants were able to successfully inhibit their responses

on half of the stop trials, striking a balance between success and failure rates. This adjustment was done using a staircase-like process.

From the data, we calculated the stop-signal reaction time (SSRT) for each participant. The SSRT represents the processing time required to inhibit a strong motor response. This was determined by subtracting the average SSD from the median reaction time observed in trials without the stop signal. This metric quantifies how quickly and effectively participants can stop themselves from responding when prompted by the stop signal.

3.2.2.3 The "Hidden-State" task Finally, the hidden-state task was used. This task was developed and elaborated on the first chapter of this thesis. The

3.3 Results

3.3.1 Reversal Learning Task

For the Reversal Learning Task, 139 subjects are shown, 69 control subjects and 70 subjects diagnosed with OCD. Most analysis are shown in comparison between these two groups as our main goal was to utilize such tasks to distinguish between groups.

The number of correct responses (Fig. 9), while slightly larger on the OCD group, is not significantly different from the control group.

The same can be said for the score obtained in the task for the 2 groups. The OCD group showed marginally non-significantly larger scores (Fig. 10).

The crucial measure reported in this task is, however, the Reversal Latency. The Reversal Latency is calculated by counting the number of

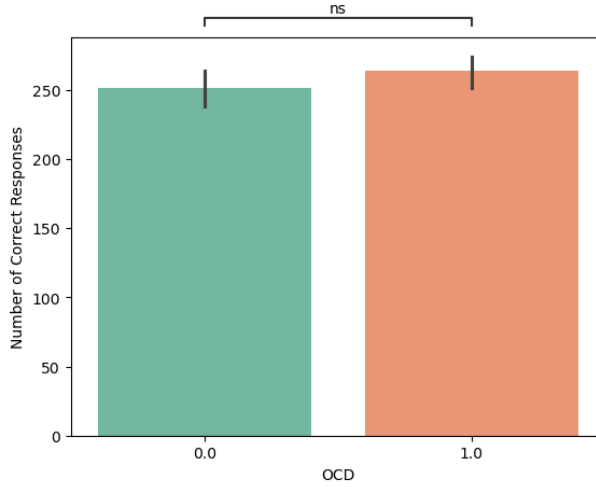


Figure 9: "Number of Correct Responses in the Reversal Learning Task"

The number of correct responses is shown for both the control (green) and OCD (orange) groups. $n=139(69/70)$, the error bar represents the 95% confidence interval.

trials in which a participant, after a reversal in the task contingencies, continues to insist before adapting to the new task parameters. In our sample, the two groups were not significantly different from each other (Fig. 11). We observed a larger spread in the data for the healthy participants and a non-significantly larger Reversal Latency.

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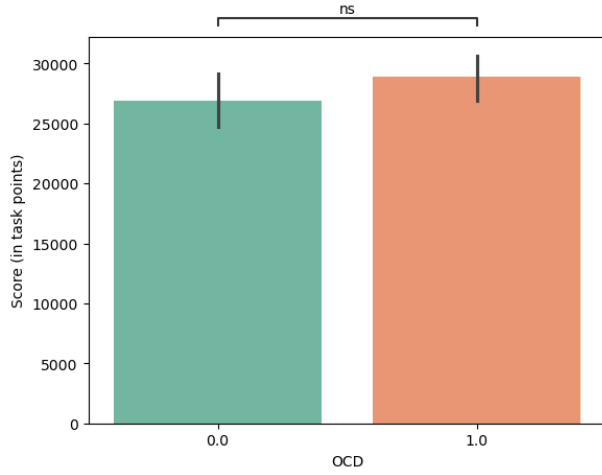


Figure 10: "Score in the Reversal Learning Task"

The score for both the control (green) and OCD (orange) groups. $n=139(69/70)$, the error bar represents the 95% confidence interval.

Our results did not go in accordance with previous literature, which leads us to ponder on the reasons why in the following discussion.

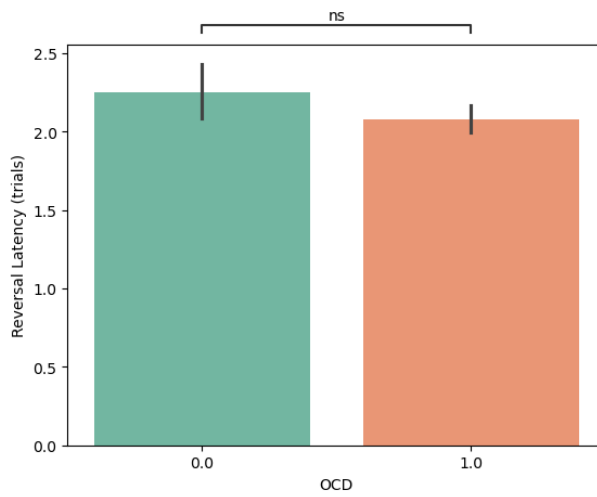


Figure 11: "Reversal Latency in the Reversal Learning Task"

The Reversal Latency for both the control (green) and OCD (orange) groups. $n=139(69/70)$, the error bar represents the 95% confidence interval.

3.3.2 Stop-Signal Reaction Time Task

In the Stop-Signal Reaction Time Task, we started by analyzing its prime metric, the Stop-Signal Reaction Time. This is calculated using a race model, which posits a competition between the "go" process (executing a response) and the "stop" process (inhibiting it). Two primary methods are commonly employed: The mean method and the integration method (based on the distribution of the data). By calculating the SSRT using the mean method we get Figure 12.

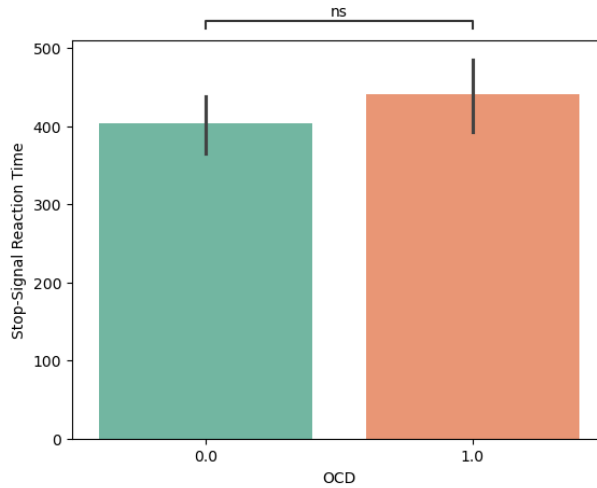


Figure 12: "Stop-Signal Reaction Time"

The Stop-Signal Reaction Time for both the control (green) and OCD (orange) groups. $n=124(62/62)$, the error bar represents the 95% confidence interval.

3.3.3 The "Hidden-State" task

3.4 Discussion

3.5 Author Contributions

3.6 Acknowledgements

4 Behaviour in the real world

4.1 Introduction

From the clinic, we venture further and into the real world. In this chapter we explore a more complex task that was ran outside the lab, its advantages and pitfalls alike.

4.1.0.1 The rise of AI The past quarter century has seen a dramatic swing in the match up between humans and artificial intelligence (AI). Today, machines reign supreme in tasks we deemed human exclusive only a few decades ago. This shift can be primarily attributed to advancements in neural network algorithms. In 1997, Gary Kasparov - the best of human chess players - was defeated by a computer program known as Deep Blue [Hsu, 2002]. Albeit incredibly powerful, this computer depended on very specific hardware and software configurations to be good at the game. Deep Blue was exclusively good at the game of chess. It contained a tree of millions of possible chess plays, a database of the most common openings and endgames and the aid of human engineers to steer it in the right direction. The definitive domination over humans in this type of task only came with the dawn of machine learning and artificial neural networks (ANNs). Neural networks are a type of algorithm inspired by the neuron-based, layered way in which biological brains compute problems. Due to their architecture, they are capable of learning from data. During training, they adjust their internal parameters (weights and biases) to adjust to the problem at hand. This training involves using large amounts of data to improve the network's performance at a given task, whether it's recognizing images, understanding language, or playing complex

games. These algorithms are capable of generalizing: they adjust to both the task at hand and, even with incomplete data (contrarily to brute-force algorithms), are able to infer what to do in a never before seen problem of the same kind.

From arcade classic Space Invaders to one of the most complex games ever played in Go, there has been a noticeable shift in prowess between human players and artificial intelligence (AI). Today, machines reign supreme in tasks we deemed human exclusive only a few decades ago. This shift can be primarily attributed to advancements in neural network algorithms. In 1997, Gary Kasparov - the best of human chess players - was defeated using the brute-force algorithm Deep Blue [Hsu, 2002]. Albeit incredibly powerful, this algorithm depended on very specific hardware and software configurations to be good at the game. Deep Blue was exclusively good at the game of chess. It contained a tree of millions of possible chess plays, a database of the most common openings and endgames and the aid of human engineers to steer it in the right direction. The definitive domination over humans in this type of tasks only came with the dawn of Machine Learning and artificial neural networks (ANNs). Neural networks are a type of algorithms inspired by the neuron-based, layered way in which biological brains compute problems. Due to their architecture, they are capable of learning from data. During training, they adjust their internal parameters (weights and biases) to adjust to the problem at hand. This training involves using large amounts of data to improve the network's performance at a given task, whether it's recognizing images, understanding language, or playing complex games. These algorithms are capable of generalizing: they adjust to both the task at hand and,

even with incomplete data (contrarily to brute-force algorithms), are able to infer what to do in a never before seen problem of the same kind.

4.1.0.2 Investigating complex skill learning Despite this, one aspect where humans continue to hold a significant edge is in the speed at which we acquire and refine our skills [Tsividis et al., 2017] [Lake et al., 2017]. Lake and colleagues pose that the differences in learning may be due to the different learning apparatuses. Complex skill learning, can be formalized as a journey through a vast space of potential strategies. It is by traversing this complex strategy space that agents eventually land on critical points that lead to task performance - i.e. they learn [Ullman et al., 2012]. While designed as reward-based gradient descent for artificial agents [Mnih et al., 2015], the exact nature of how humans navigate this space remains uncovered. One position is that it follows a similar process. Alternatively, some pose this process is influenced by rational inference and priors [Tsividis et al., 2017] [Martin-Maroto and de Polavieja, 2018]. Given the wealth of theories, a significant challenge in this field of research remains that games serve as excellent benchmarks for assessing intelligent search strategies and decision-making processes yet, they are not inherently designed to reverse-engineer complex skill learning. If we are to draw parallelism between different types of learning in different agents, we ought to investigate tasks that are designed to investigate underlying cognitive processes - i.e. learning.

4.1.0.3 Is it smarter than a third grader? To answer any questions about how intelligent an agent is, we must first begin to

agree on a definition of intelligence. AI’s definition has often landed on benchmarking measures of skill [Chollet, 2019]. Skill is heavily dependent on priors. With unlimited experience, one could train a system to become of an arbitrary skill level. Cognitive sciences have long gravitated towards definitions that more closely relate to one’s capacity to generalize: and that is indeed the promise of ANNs. The ideal intelligent machine is such that, when faced with the same stimuli a human has experienced in a task, produces a trajectory through policy space that matches that of the humans [van Opheusden and Ma, 2019].

4.1.0.4 Not too simple, not too complex On the opposite end of the spectrum of complexity, the tools of psychophysics have been instrumental in many of the great findings in cognitive psychology. The field of psychophysics examines the fundamental relationships between physical stimuli and their cognitive-behavioral reactions. One family of widely studied tasks in this domain is that of alternative forced choice (AFC) tasks. Not unlike the tasks presented in the previous chapters of this work, subjects are oft asked to make actions in which some outcomes lead to reward. For example, in a simple sensory discrimination task, a subject may be asked whether they see blue color on a screen. Plotting responses against the intensity of blue hue generates a so called psychometric curve. In it, we would see that the proportion of correct detections would increase with hue. This observation allows us to draw a direct relationship between the task and behaviour. With complex tasks, this relationship is much harder to establish. Moving a rook or a knight in a game of chess is not easily relatable to any task parameter.

The dichotomy presented has led to a great divide in the complexity of tasks studied by the fields of AI and of Cognitive Sciences. A way to quantify complexity in a task is by observing the dimensions of its state-space - that is, the number of possible different states that an agent can reach in any possible sequence of actions.

Decision trees - i.e. schematic representations of the state-space - are a powerful way to model planning tasks [van Opheusden and Ma, 2019]. While admittedly reductionist - as it ignores things like task difficulty and whether there is uncertainty in the task - by calculating the dimensions of a decision tree we can derive a measure of task complexity. van Opheusden and Ji Ma (2019) compare several families of tasks as to their state-space complexities. They posit that there is a big gap in task complexity between the two fields and leave us, cognitive scientists, with the challenge to venture away from simplicity and embrace complexity in order to align human and machine intelligence.

4.1.0.5 Embracing complexity To bridge this gap, we have developed a novel game specifically tailored to explore and illuminate the mechanisms of human skill learning. This game, unlike traditional ones, is uniquely engineered to delve into the depths of how humans learn, adapt, and master various strategies and skills. It's not just a platform for challenging human intellect against artificial intelligence but a tool to dissect and understand their behaviour as well. Our game aims to provide valuable insights into skill acquisition, offering novel a perspective on how humans adapt and excel in complex decision-making environments.

4.2 Materials and Methods

4.2.1 hexxed

To investigate how complex skill learning ensues, the right tool is needed. After exploring the landscape of possible tasks^{4.1}, we developed a novel task: *hexxed*. *hexxed* is a puzzle-like videogame task that challenges it’s solvers to explore and solve complex problems. Solving *hexxed* requires carefully planning moves ahead in both space and time.

We seek to combine the strengths of simple and complex tasks. This led us to identify several design constraints. First, the “state space” of the task is sufficiently large, motivating subjects to develop efficient strategies to solve it. Second, the values of all states and actions should be pre-defined so that the value of a player’s choice is unambiguous. Third, stimuli can be generated programmatically, allowing us to comprehensively sample subjects’ changing strategy across stimuli throughout learning. Fourth, the task progresses in difficulty, so that it can be used to observe skill formation within individuals. Finally, the task generates enough data, within and across subjects, to sufficiently sample the distribution of strategies that subjects employ. Over multiple iterations we arrived at a puzzle-based video game that meets these constraints: hexxed.

hexxed can be seen as a time-dependent version of the traveling salesman problem [Gavish and Graves, 1978]. This class of problems requires combinatorial optimization, in that finding the optimal solution can require evaluating all possible solutions (known as a “brute force” search), which grows exponentially with the number of targets.

Furthermore, the number of possible puzzles grows exponentially with the number of targets used. At the same time, both puzzle and solution spaces are discrete and enumerable. These properties position *hexxed* as an ideal psychophysical tool for probing how humans solve a complex problem.

4.2.2 The task

While *hexxed* takes the shape of a game, it is designed first-and-foremost as a psychophysics task.

The game is interacted with by touching on a touch-sensitive device.

When first presented with it, the players are immediately shown the first puzzle with no further instructions other than the displayed on-screen: "controls: tap and swipe".

In *hexxed*, the goal is to rotate a collector (which is the player's controllable component) around lobes of a hexagon and collect rewards. Rewards are administered when targets are collected. Targets can spawn in any of the lobes of the hexagon (with a minimum of 1 target in total and a maximum of 1 target per lobe).

In *hexxed*, time is discretized. For every action a player takes, time progresses one unit into the future. This means that *hexxed* has no inherent time-pressure to be solved and that it is the players who determine the pace at which they solve it.

At any point, players can make one of two actions. To tap or swipe in any of the 6 lobes of the hexagon, for a total of 12 possible actions.

Tapping on any of the lobes causes the collector (i.e. the player) to move there. The collector only moves one lobe at a time. If the player clicks the lobe adjacent to them, they move there and progress

1 unit of time. If the player clicks a lobe that is non-adjacent to their position, time progresses the corresponding distance (e.g. a player clicks the lobe on the opposite side of the hexagon - 3-away - and time progresses 3 units). Whenever time progresses, targets grow in size. Targets of size 1, are worth the least possible reward, whereas targets of size 6 are worth maximum reward. Targets of size 7 expire and are no longer visible or valuable to the player. Swiping on a lobe without a target is treated as a tap. Swiping on a target collects points corresponding to the target's size.

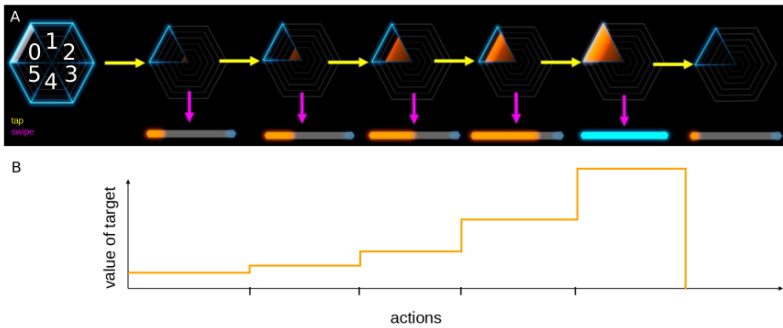


Figure 13: Game play progression in response to player actions

A) Example trial for a single-target puzzle (level 1). In each moment, one of two actions can be made: a tap (as illustrated by the yellow arrow) or a swipe (pink arrow). In this example, actions are restricted to the lobe containing the target. A tap leads to an increase in size, while a swipe leads to a collection event. Following collection, the player earns points. B) The value of targets is defined as the square of their size. After 6 non-collecting actions, the value of the target drops to 0.

The task is then to navigate the hexagon's lobes and collect targets

as large as possible.

The task is designed as an np-complete problem. Np-completeness (or nondeterministic polynomial time completeness) refers to when the solution for any possible problem in a task can be verified in polynomial time - using a brute-force search (i.e. an algorithm that attempts all possible solutions). This is a crucial feature of *hexxed*, as it allows us to arbitrarily manipulate the complexity of any puzzle while always knowing the optimal solution is within computational reach. Np-completeness grows the posed challenge exponentially, giving more dynamic range for testing the limits of performance. Simultaneously, the posed problem becomes isomorphic with a large class of well-studied problems.

Complexity in *hexxed* can be manipulated in two ways: 1) by changing the number of targets on screen and 2) by changing the order in which targets are spawned. A growing number of targets forces the player to move more around the hexagon to collect them, whereas the order in which targets are spawned influences how those movements are made.

The task is comprised of 6 different levels of difficulty - corresponding to the number of targets displayed on screen for that level. In the first 4 levels, the players are probed with all the possible puzzles (combinations of spawn locations and their rotational variants). In levels 5 and 6, only half the possible puzzles are shown. To progress in the task, players must approximate optimality to a varying degree (e.g. for level 1, players can lose no more than 15 points in order to complete the level).

4.2.3 Scoring

Individual targets grow up to six times and are worth n^2 points: from 1^2 - as they spawn - to 6^2 points - at their ripest. The exponential nature of target value places weight on collecting the target as large as possible. However - given that the game is self-paced - too many interactions with the screen will exhaust a target and lead to 0 reward (see 4.2.2 for the task design).

4.2.4 Action-sequences

Our main object of analysis is the action sequence: the series of taps and swipes that a player uses in attempting to solve a single puzzle. In our task, several reference frames are possible when encoding action sequences. 1) From the perspective of the game board ('world-centered'); 2) from the perspective of the target ('target-centered'); 3) from the perspective of the collector ('agent-centered'). To determine which of these most likely represent the player's reference frame, we measured the log likelihood of player actions according to each frame. By this metric, the 3rd frame will be shown as most likely. To encode action sequences, we number the 6 lobes of the hexagon from 0 to 5 clockwise, relative to the collector's location (Fig. 1A). Since the collector has no location at the puzzle start, we use target-centered coordinates for the first action. We then encode the sequence of player taps or swipes on these lobes as a list of actions. Taps are encoded as the number of the lobe that was pressed and swipes are encoded as 6 more than the number of the lobe pressed.

In the example action sequence for level 1 (Fig. 1C), a subject performed a tap on the lobe to the left of the lobe which had a target

(5) ; then, the player tapped on the lobe with the target for 5 times - on the first time, the slider moves one to the right (1); then stays four times on the same lobe (0,0,0,0); finally, the player swiped on the lobe with the target (6) to get the maximal reward. This action sequence is encoded as [5,1,0,0,0,0,6].

4.2.5 Optimality in hexxed

Due to *hexxed*'s design, we are capable of calculating the optimal solution for any given puzzle.

To calculate the optimal course of actions, we trace which action-sequences lead to the maximum possible score for a given puzzle (see figure 15).

Because of the nature of the game design, in some circumstances, several action-sequences may be equally optimal. As the puzzle complexity increases, however, the fraction of optimal action-sequences becomes smaller.

For level 1, optimality is trivial since a single target can always be collected at maximum value: Any combination of moves that ends at the target lobe after 6 actions allows for maximal point collection.

For level 2, two targets are available for collection. However, depending on whether targets are in adjacent lobes or not, they may or not be both collectible at maximum size. This means that, on non-adjacent targets, the player must sacrifice points of the first target to collect the second at its ripest. When targets are n lobes away from each other, the maximum possible score is to collect the first target at step $6 - n + 1$ so that there are enough moves available to rotate n lobes and collect the second target at its maximum value (e.g. for two

targets that are 3 lobes away from each other, the maximum ripeness is of $(6 - 2) + 6 = 10$ or $(6 - 2)^2 + 6^2 = 52$ in points.

For levels 2 through 6, two competing heuristics can be used to maximize reward efficiently: targets can be collected a) in order of their appearance or b) in order of their proximity (see figure 13). As puzzle complexity increases to levels 3 and onward, collecting targets in their order of appearance is no longer the optimal strategy.

The way *hexxed* is designed, thus, forces players to constantly rethink and revisit their previously optimal policies. By virtue of its NP-completeness, *hexxed* requires players to update their policies to incorporate progressively higher-order dependencies, making it a great candidate to investigate complex skill learning.

4.2.6 The game

hexxed is instantiated a game. Unlike typical lab-based research tasks, *hexxed* was developed using a game engine. Game engines are a family of software that allows for quick and precise manipulation of both algorithmic logic and graphics. *hexxed* was developed using *Godot* [Contributors, 2023], an open-source game engine chosen for its openness and flexibility.

In the process of coding *hexxed*, we leveraged its algorithmic nature and made it entirely generative - The game does not consist of static images and illustrations, instead, the graphics are generated on the device’s graphics card. This allows for an extremely small executable size: the entire game is only $\sim 3.5\text{mb}$ in size. The game is available on both android and iOS devices and, due to its small size and speed, the game is available on an estimated 95% of all smart devices.

All of *hexxed*'s code is open-source and available under a Creative Commons license at <https://hexxed.io>.

4.2.7 Data collection and recruitment

The data collection process for the *hexxed* experiment encompassed both laboratory and online environments.

In the controlled laboratory setting, study participants engaged with the game on a large touch-sensitive screen located within a quiet, controlled environment. The participant pool consisted of individuals from the Champalimaud Research community, comprising undergraduate students, technicians, graduate students, and postgraduates. It is important to note that unless otherwise specified, the *hexxed* dataset presented in this dissertation primarily stems from online data collection.

Conducting data collection online introduced unique challenges, notably those related to scalability and the breadth of data collection. By framing the experiment as a game, we anticipated data to be generated at varying and unpredictable rates, as the game was accessible to anyone, anywhere, at any time. To maximize participation, we collaborated with the science communication offices of both Champalimaud Research and the University of California, Berkeley, resulting in extensive media coverage that attracted an initial 8k participants. Consequently, we had to develop tailored solutions for data formatting, storage, and transfer to effectively manage the influx of data.

It is essential to emphasize that, despite its gamified presentation, *hexxed* is a scientific experiment. Therefore, we diligently adhered to the ethical and deontological guidelines set forth by the Champalimaud

Research ethics committee. This encompassed ensuring compliance with the General Data Protection Regulation (GDPR), obtaining informed consent from participants, ensuring data anonymization, and providing participants with the option to opt out, among other ethical considerations. The comprehensive license agreement and privacy policy governing data usage for *hexxed* can be accessed at the following URL: <https://hexxed.io>. Additionally, any ethical approval documentation and related ethics materials can be made available upon request through the same URL: <https://hexxed.io>.

4.2.8 Data Architecture

To manage the data generated by *hexxed*, we developed a customized data format implemented within a JSON structure. The data handling process involves the *hexxed* executable, which generates a file that logs every interaction players have with the screen during a playthrough. Each interaction, encompassing actions such as touches or swipes, results in the creation of a list containing pertinent data related to that specific interaction. These lists are dynamically generated and continually appended to memory as the game progresses.

At the culmination of a playthrough, which occurs either when a player completes or fails a level or when they exit the game, all the accumulated lists stored in memory are encoded into a JSON payload. This payload is subsequently transmitted to a secure server designated for this purpose.

Upon receipt at the server-side, the incoming payload is archived within an SQL database specifically designed to store data from all playthroughs across all players. This database serves as a comprehen-

sive repository containing data from all facets of the game.

For a detailed illustration and documentation of the payload structure, an example is provided in Supplementary Table 5. This table offers insight into the format and contents of the JSON payload, aiding in the comprehension and analysis of the data collected during *hexxed* playthroughs.

4.2.9 Data Processing and Analysis

Due to the size and complexity of the collected data, we had to build a custom pipeline for processing and analysis. All of the python pipeline scripts and documentation can be found on *hexxed*'s github repository at <https://hexxed.io>.

After downloaded, the data is extracted from the JSON payload into a pandas dataframe. This process requires careful consideration, as it is easy to underestimate the needed computational resources when dealing with such a large dataset.

Raw data is saved in a compressed format. Now, its efficient parsing becomes the challenge. Because of the payload structure, to extract the data into an analysis-friendly format we naively would have to iterate through thousands of rows, parsing each JSON string as appropriately *typed* columns in a dataframe. Luckily, python and pandas allow for alternatives to this iterative process. Using the split-apply-combine [Wickham, 2011] framework, and after thorough optimization steps, we transform our entire dataset into a large long-formatted pandas dataframe (i.e. a timeseries). Following the preprocessing steps, the data is now easily *query-able* and manipulable. Our final dataframe has a coarse-ness of one action per row - the highest granularity the raw

data allows for. Using pandas groupby, it is easy to adjust on which variable we want to group this dataframe, allowing for efficient analysis of any type. All of these steps took our data processing pipeline from approximately two hours to no more than two minutes.

While shortly described, we found this process to be both extremely valuable and time consuming. As the scope and dimensionality of datasets increases in behavioral sciences, it is increasingly important that scientists adopt novel strategies with which to handle their data. Efficient data handling and transparency allow for replicability and 3rd party analysis/criticism/expansion, and thus, all our processing and analysis scripts are available at <https://hexxed.io> upon request.

4.2.10 Scope of the current work

By interviewing our human subjects during the testing phase, we learned that they learned to play this game in two related but distinct phases: 1) Learning the rules and objectives of the game – Since subjects were not given any verbal instructions, they must search for sufficiently rewarding action sequences. In the first level, both humans and AIs need to learn to collect a single object at its full size. This requires choosing from over 4,000 viable action sequences. Interviews indicated that rule learning process largely (but not completely) takes place in level 1. 2) Knowing the rules and objectives, identifying the best action – once subjects understand the rules, they must make an inference regarding how to apply these rules to maximize reward. Interviews indicated that this process starts in level two and onwards, when subjects must determine an optimal path to collect two or more objects (more directly comparable to the combinatoric optimization

of the traveling salesman problem). We have limited the scope of our current analysis to the first of these two phases, because formally it corresponds to the problem with the smallest search space, and because understanding this phase should help constrain our models of the second phase.

4.2.11 Comparison to null models

To compare human behavior to that of artificial agents, we needed to encode the game into an API that could be called by the agent. For this, we chose OpenAI’s Gym, which offers an API for communication between artificial agents and a task of interest. Several agents were used in our task. 1) 0th order agent - assumes a uniform distribution of possible actions. This agent draws a random number from the possible values in our action-space (i.e. an action between 0 and 11). 2) 1st order agent - selects actions based on the observed probabilities of user actions (which is non-uniform). 3) 2nd order agent - selects actions based on the observed probabilities of user actions conditioned on users’ previously chosen action. For each of these agents, we calculated their log-likelihood in explaining the data using 5-fold cross-validation.

4.2.12 Beyond this thesis

The hexxed dataset has underwent several iterations that were left beyond the scope of this work. Amongst this rich dataset, we have in-lab participants from whom pupil tracking data was collected.

4.3 Results

4.3.1 Introduction

hexxed was designed to explore complexity in learning and, as such, learning comprises a large part of this work. In fact, the scope of this thesis will solely be on the phase where subjects learn the rules of the game - level 1.

4.3.2 "Picky" - Humans explore the space of possible rewards unevenly

hexxed's dataset is vast. Let us start with a low level observation: the first ever interaction with the task.

As the game provides the player with little to no instructions (see 4.2.2), how and where the players interact with the screen can give insight on whether, in broad strokes, the game is intuitive or, at the very least, interpretable.

In Figure 14, the first ever interaction with the game is shown. Players seem to have a common *prior* that they ought to press the hexagon on the lobe that is highlighted. That lobe is where a target becomes available for collection, thus, pressing it is a sensible option.

In future analysis, we abstract away the coordinates of each action, using instead a discrete action-based encoding (see methods).

Our task can be formalized as a Markov Decision Problem (MDP) (Figure 15). As continuous choices are made, specific strategies or ways of playing emerge. In Figure 16, a flattened version of the MDP of Figure 15 is populated with the action-transitions made by our subjects. We visualize how humans as a population traverse this

MDP in their first puzzle attempts. Notably, the path constituting interactions with the target-containing lobe (top horizontal path) is overrepresented.

As players explore level 1, they eventually collect points. In Figure 17, we show the probability of our subjects, on a given trial, getting any of the seven possible reward sizes. Rewards of 0 are the most common - this reflects that failures are the most common outcome in the task. Looking at the remaining reward sizes, the second most common collected reward is the maximum reward - a requirement for optimal level completion. The only other reward size that is differently explored is the smallest reward possible.

We next modeled the observed distribution of human action sequences using a series of generative models of growing complexity.

The 0th order agent selects uniformly among the 12 possible actions. The 1st order agent selects actions determined by the observed probabilities of human actions (Figure 18). Finally, the 2nd order agent introduces a momentum prior by considering the previous action.

We then calculated the cross-validated log-likelihoods for all three models. We find that, as we increase the complexity of the agent, we gain more explanatory power over the data (Figure 2E). This shows that humans prefer smoothly changing trajectories that take them to a target of interest.

In order to progress in the game, sequences of actions are made. Our players performed 14158 unique action sequences. In Figure 20, the 20 most common are shown - the encoding is explained on chapter 4.2.4.

We see that the most common action is a single swipe on the

correct lobe - which leads to a reward of 1. The second most common is one of the optimal solutions - tapping only the lobe where reward is available and swiping at the last possible moment. The third does not include swipes, and thus leads to a reward of 0. The fifth and seventh represent, respectively, navigating the hexagon clockwise and counterclockwise before swiping at the last possible step - both leading to optimal reward.

Our subjects explore the space of possible rewards unevenly.



Figure 14: "First touch"

A heat-map of the participants' first ever touch coordinates on the screen, superimposed with a screenshot of the game. Player touches are largely centered around the gray bar on the bottom. Unbeknownst to the player, this bar is an indicator that a target will appear on that lobe.

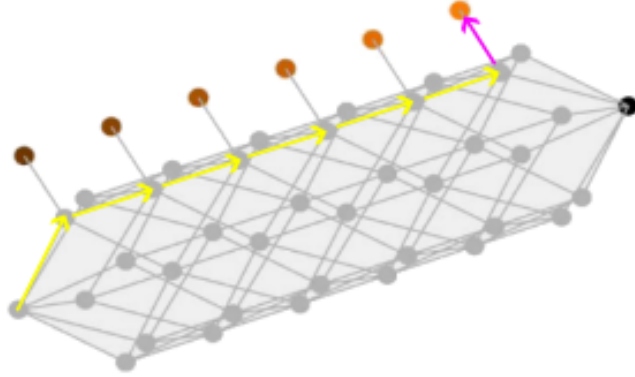


Figure 15: Schematic of a single target puzzle as a Markov Decision Process

A single-target puzzle represented as a Markov Decision Process (MDP). From the initial node, each player action propagates them 1 step rightwards. Lines show deterministic transitions resulting from specific actions. The high-contrast circles denote terminal states, with brighter orange corresponding to larger reward and black symbolizing a reward of 0. Arrows denote an example optimal solution for a puzzle of the kind of panel, yellow indicating taps and pink indicating a swipe.

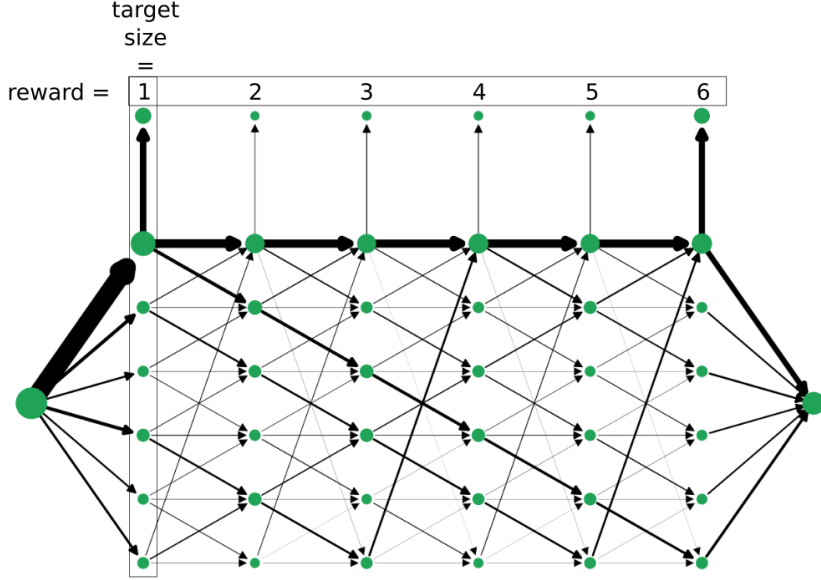


Figure 16: Action transitions in level 1 puzzles in a flattened version of Figure 15.

Frequency of observed action sequences on the flattened MDP. All subjects start on the leftmost node and traverse the graph rightwards. Line widths represent the occurrence of that state transition within the data set (thicker lines indicate higher probability). Node sizes represent the frequency of their occurrence. Rewards are available on the nodes on the top row, growing in value from left to right. States are numbered in target-centered coordinates, as agent-centered coordinates are not compatible with this visualization. The rightmost node represents a reward of 0, achieved when a target exceeds the size of the board. Columns distinguish states in which the target is of a certain size. Rows represent the position of the collector relative to the target (n=3994).

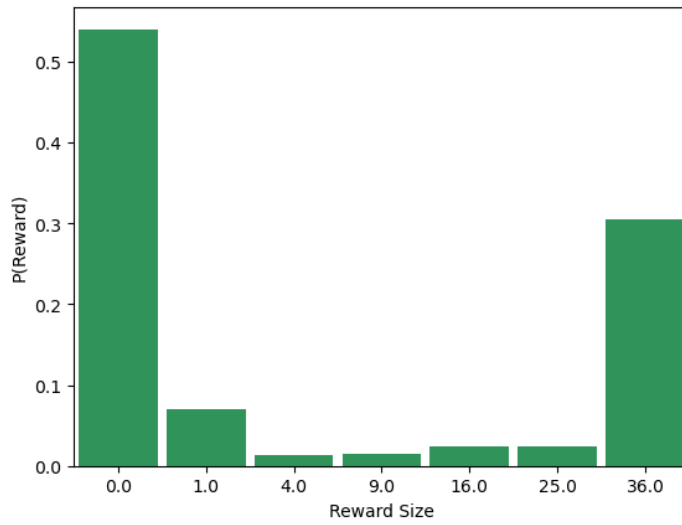


Figure 17: The probability of getting rewards for all possible reward sizes.

The probability of getting rewards for all possible reward sizes in the first time that subjects attempt the puzzle (n=3994).

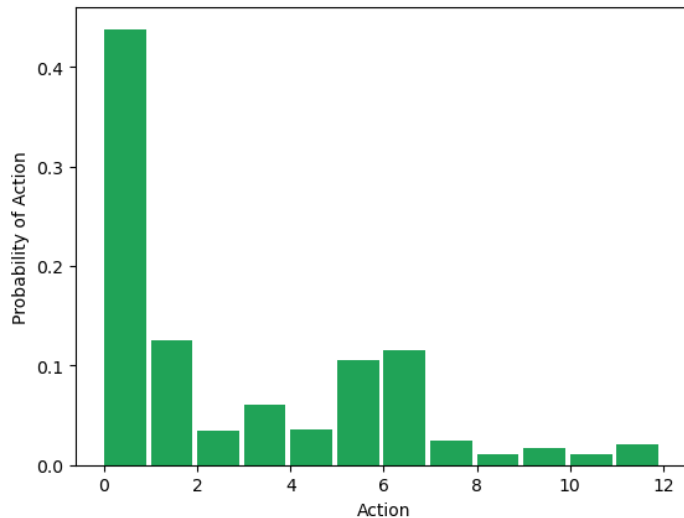


Figure 18: The probability of actions

The probability of each action being performed by our subjects. The most common action is “0” - tapping on the lobe where the target is present. The other common actions are “1” and “5”, representing clockwise and counterclockwise navigation, and “6”, representing an in-place swipe (n=3994).

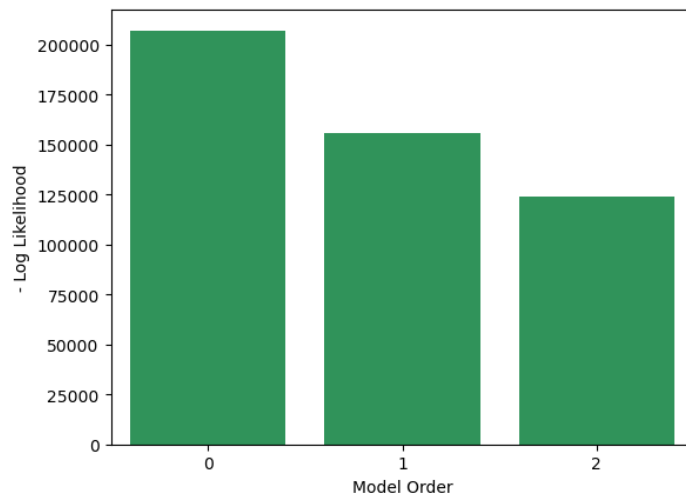


Figure 19: The negative log likelihood for all three models
The negative log-likelihood (5-fold cross-validated, lower represents a better fit) for all three models.

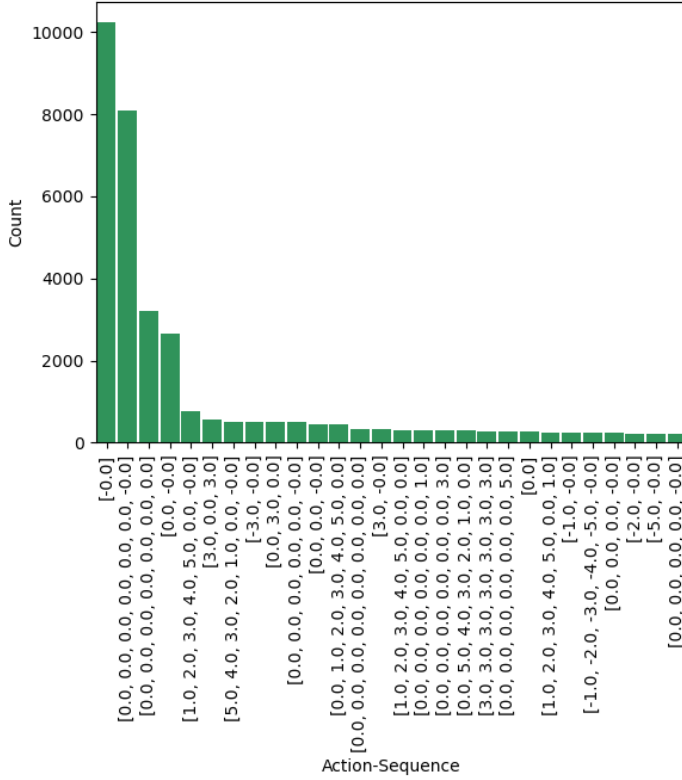


Figure 20: The most common action sequences

The top 20 action sequences and respective frequencies of occurrence - in total occurrences across all players. There are a total of 14158 unique action sequences. An action sequence [6] represents a collection event leading to a reward of size 1. The second most common action sequence “[0, 0, 0, 0, 0, 0, 6]” indicates an optimal collection event leading to maximal reward. The fifth and seventh most common action-sequences represent navigating the lobes clockwise or counterclockwise and then collecting reward at its maximum size.

4.3.3 "Sticky" - Humans are persistent in their learning strategies

Having established the prior knowledge that humans bring to the task, we now examine how this knowledge is modified across trials with experience. In Figure 21, we plot the conditional probability of getting reward r in trial t given the reward in the previous trial $t-1$. We observe that the edges of the diagonal seem overrepresented. This is consistent with the points raised on 4.3.2. Rewards of 0 (i.e. failures) lead to revisiting failures, whereas optimal rewards lead to optimal rewards once more. The center of the heat-map is more fuzzy. Intermediate rewards suggest intermediate sub-optimal strategies where perhaps subjects have already learned the basic controls of the game, however, are still not capable of solving it optimally.

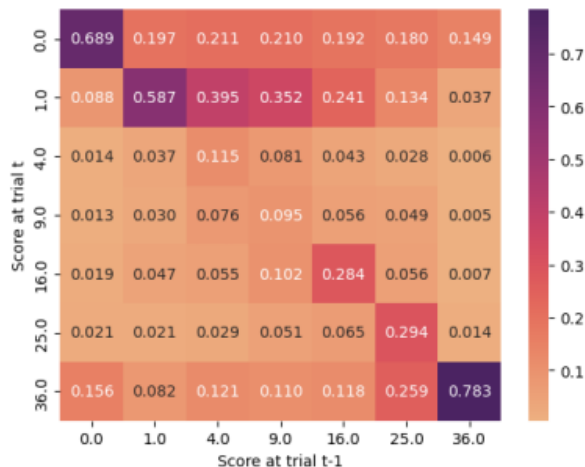


Figure 21: Reward Transition Matrix

The probability of getting a reward r on trial t conditional on the reward of the previous trial $t-1$: $P(r_t|r_{t-1})$. Darker colors indicate higher probability. Columns of probability sum to 1. (n = 3994)

One limitation of this depiction is that multiple action sequences can lead to the same reward. To address that, we look at the action sequence transitions from sequence s on trial t from the sequence done on the previous trial $t-1$ (Figure 22). Again, the diagonal appears to be overrepresented. This signifies that players often repeat the same action sequence they performed before. This is especially the case for optimal solutions (e.g. sequences 2, 5 and 7), where it makes most sense to keep repeating ones current action sequence. Another feature is the asymmetry between the lower and upper sides of the diagonal. Action sequences are most commonly taken from least common to most common, perhaps signifying some conversion towards learning.

Our subjects are persistent in which actions they perform, both when looking at which reward size they explore throughout learning as well as when looking at specific action sequences.

4.3.4 "Leapy"

We finally zoom-out and observe the entire progression of a player throughout a level. In Figure 23, each individual player is represented by a randomly colored line. To complete level 1, subjects must collect a certain amount of reward (see 4.2.2). Here, this threshold is represented by the dashed line. In these traces we observe a pattern of low performance on a number of sessions followed by a rapid leap that leads to success. In the top panel, a histogram of when players reached the reward threshold. We see an early centered yet long tailed distribution. This hints that albeit fast at learning, subjects tend to do so in a leapy fashion - consistently failing for several attempts until finally achieving the threshold in a single final attempt.

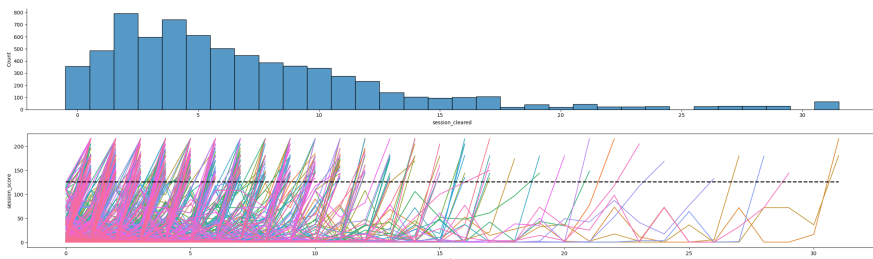


Figure 23: Performance during level attempts

Top panel: Summary visualization of the bottom panel. Histogram of number of attempts needed to complete level 1 reveals that many subjects solve the task rapidly, while a few take significantly longer, leading to a right-tailed distribution. Bottom panel: Individual traces of performance across level attempts during learning (i.e. until level completion). To complete a level, subjects must reach a fixed reward threshold, here represented as a dashed line. Hue is randomly assigned for easier visual discrimination across individual traces.

4.4 Discussion

Embracing complex large datasets comes with a myriad of challenges. Our work highlights the importance of careful data collection and manipulation. We leveraged the wide dissemination of smartphones and unlocked their potential by launching our behavioral task as a downloadable app. The outcome of this was a rich and large dataset that allowed us to explore the intricacies of human complex learning.

Our work focused on the early stages of learning. Crucially, our task comes with little to no instructions - this gives us first-hand insight into the process of learning the rules and mechanics of the game. One of the interesting findings of our work is the first ever action performed on the game. We showed that subjects largely tap on the coordinates congruent with the future appearance of a target. This could be due to the presentation of a hint on screen that highlights where a target is going to appear. It hints that our players follow some sort of shared prior, going in stark contrast to what an artificial agent would do, for example, when solving this game.

We then demonstrated that subjects do not evenly explore the space of possible actions in the task. They instead focus on the highest possible rewards and the actions that most directly lead to them. Simple actions of tapping in place where rewards are available were preferred.

What the most parsimonious solutions are was revealed by how our subjects navigated the puzzles. We found that solutions that involve the least understanding of the game’s rotational properties were more appealing to our subjects. Subjects preferred sequences of actions that directly led to reward. The exceptions to this were sequences

that rotated clockwise or anti-clockwise on the game board before collecting optimal reward - this strategy is valid for level 1 because the amount of steps required to grow a target optimally are the same as traversing the whole board in one of the rotational orders (i.e. 6 steps). Nonetheless, the top most performed action-sequences are generally ones that follow simple heuristics. This suggests that our subjects share these simple heuristics about rotational properties and prefer them over more complex yet equally rewarding action-sequences. We deemed this phenomenon pickiness, as subjects tend to choose specific routes in which to navigate the state-space. Interestingly, we demonstrated that models growing in complexity increasingly approximated pickiness. Future research is needed to assert whether an ever more complex agent could fully replicate pickiness, but it is safe to say that our results point towards a model that takes into account priors.

We then looked at phenomena happening over trials. Both in reward and action space, our data was heavily centered around repeating actions over trials. Moreover, these spaces were not explored evenly. Our subjects tended to go from the less to the more optimal. Action-sequences thus took a directionality congruent with learning.

When observing the learning trajectories, we find that our subjects follow a pattern of fast yet non-gradual learning. Instead, their performance remains low until a point where it ballistically grows into clearing the level. This is what we describe as “leapy”. A naive agent performing bottom-up learning would follow gradient ascent. Instead, our subjects show signatures of top-down learning. A process by which humans converge on common explanatory solutions which they replace only upon collecting sufficient evidence to the contrary. This solidifies

the reliance on priors to perform in our task.

4.5 Author Contributions

- Task design and implementation by T. Quendera and G. Agarwal.
- Analysis and discussion by T. Quendera.

4.6 Acknowledgments

I would like to acknowledge the contribution of Gautam Agarwal towards the work developed in this chapter. All of it was done in close collaboration with him. None of it would have been possible without him and our seemingly infinite hours of discussion. Thank you.

5 General Discussion

5.1 Brief summary of the main findings

5.2 Opportunities for expansion

5.3 Concluding remarks

6 Supplementary

device-ID	session-start	time	sw-flip	sw-offset	sw-subwave-num	sw-number	mo-y	mo-x	mo-fake-release	mo-act-drag	mo-lobe	level	version
0	0012e848890b5d222382	531464	False	4.000000	2.000000	-	121.1111099	196.539368	0.000000	False	5.000000	1.000000	2.000000
1	0012e848890b5d222382	533139	False	4.000000	2.000000	-	82.777802	-	0.000000	False	2.000000	1.000000	2.000000
2	0012e848890b5d222382	534881	False	4.000000	2.000000	-	87.222214	103.195862	0.000000	False	5.000000	1.000000	2.000000
3	0012e848890b5d222382	540839	False	4.000000	5.000000	127.222229	202.651123	202.651123	0.000000	False	6.000000	1.000000	2.000000
4	0012e848890b5d222382	542457	False	4.000000	5.000000	246.111145	10.149719	10.149719	0.000000	False	1.000000	1.000000	2.000000

Variable Name	Size	Description	Example
drone-play	(1)	time at which the background drone music restarts the loop	[374062]
mo-act-drag	(13)	0 for tap, 1 for drag (swipe).	[0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1]
mo-fake-release	(13)	multiple releases by a single press as registered by the device.	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
device-video-driver	(1)		0

Table 5: *hexred* raw data structure example

References

- [Aron et al., 2003] Aron, A. R., Fletcher, P. C., Bullmore, E. T., Sahakian, B. J., and Robbins, T. W. (2003). Stop-signal inhibition disrupted by damage to right inferior frontal gyrus in humans. *Nature neuroscience*, 6(2):115–116.
- [Asahina et al., 2022] Asahina, K., De Bivort, B., Grunwald Kadow, I., and Yapici, N. (2022). Editorial: revisiting behavioral variability: what it reveals about neural circuit structure and function. *front behav neurosci* 16: 1–3.
- [Behrens et al., 2007] Behrens, T. E., Woolrich, M. W., Walton, M. E., and Rushworth, M. F. (2007). Learning the value of information in an uncertain world. *Nature neuroscience*, 10(9):1214–1221.
- [Bellman et al., 1957] Bellman, R., Corporation, R., and Collection, K. M. R. (1957). *Dynamic Programming*. Rand Corporation research study. Princeton University Press.
- [Charvet et al., 2025] Charvet, C. J., de Sousa, A. A., and Vassilopoulos, T. (2025). Translating time: Challenges, progress, and future directions. *Brain Research Bulletin*, page 111212.
- [Chollet, 2019] Chollet, F. (2019). On the measure of intelligence. *arXiv preprint arXiv:1911.01547*.
- [Cohen et al., 2007] Cohen, J. D., McClure, S. M., and Yu, A. J. (2007). Should i stay or should i go? how the human brain manages the trade-off between exploitation and exploration. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362(1481):933–942.

- [Contributors, 2023] Contributors, G. E. (2014–2023). Godot engine. <https://godotengine.org/>.
- [Cruz et al., 2022] Cruz, B. F., Guiomar, G., Soares, S., Motiwala, A., Machens, C. K., and Paton, J. J. (2022). Action suppression reveals opponent parallel control via striatal circuits. *Nature*, 607(7919):521–526.
- [Daw et al., 2006] Daw, N. D., O’doherly, J. P., Dayan, P., Seymour, B., and Dolan, R. J. (2006). Cortical substrates for exploratory decisions in humans. *Nature*, 441(7095):876–879.
- [Ericsson et al., 1993] Ericsson, K. A., Krampe, R. T., and Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological review*, 100(3):363.
- [Faure et al., 2022] Faure, P., Fayad, S. L., Solié, C., and Reynolds, L. M. (2022). Social determinants of inter-individual variability and vulnerability: the role of dopamine. *Frontiers in Behavioral Neuroscience*, 16:836343.
- [Fechner, 1948] Fechner, G. T. (1948). Elements of psychophysics, 1860.
- [Gavish and Graves, 1978] Gavish, B. and Graves, S. C. (1978). The travelling salesman problem and related problems.
- [Gershman, 2018] Gershman, S. J. (2018). Deconstructing the human algorithms for exploration. *Cognition*, 173:34–42.
- [Gomez-Marin and Ghazanfar, 2019] Gomez-Marin, A. and Ghazanfar, A. A. (2019). The life of behavior. *Neuron*, 104(1):25–36.

- [Guo et al., 2025] Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., et al. (2025). Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.
- [Hageter et al., 2021] Hageter, J., Waalkes, M., Starkey, J., Copeland, H., Price, H., Bays, L., Showman, C., Lavery, S., Bergeron, S. A., and Horstick, E. J. (2021). Environmental and molecular modulation of motor individuality in larval zebrafish. *Frontiers in Behavioral Neuroscience*, 15:777778.
- [Hills et al., 2015] Hills, T. T., Todd, P. M., Lazer, D., Redish, A. D., and Couzin, I. D. (2015). Exploration versus exploitation in space, mind, and society. *Trends in cognitive sciences*, 19(1):46–54.
- [Hiser and Koenigs, 2018] Hiser, J. and Koenigs, M. (2018). The multifaceted role of the ventromedial prefrontal cortex in emotion, decision making, social cognition, and psychopathology. *Biological psychiatry*, 83(8):638–647.
- [Hsu, 2002] Hsu, F.-h. (2002). *Behind Deep Blue: Building the Computer that Defeated the World Chess Champion*. Princeton Paperbacks. Princeton University Press.
- [Krakauer et al., 2017] Krakauer, J. W., Ghazanfar, A. A., Gomez-Marín, A., MacIver, M. A., and Poeppel, D. (2017). Neuroscience needs behavior: correcting a reductionist bias. *Neuron*, 93(3):480–490.

- [Lake et al., 2017] Lake, B. M., Ullman, T. D., Tenenbaum, J. B., and Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and brain sciences*, 40:e253.
- [Martin-Maroto and de Polavieja, 2018] Martin-Maroto, F. and de Polavieja, G. G. (2018). Algebraic machine learning. *CoRR*, abs/1803.05252.
- [Mesulam, 1998] Mesulam, M.-M. (1998). From sensation to cognition. *Brain: a journal of neurology*, 121(6):1013–1052.
- [Mnih et al., 2015] Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., et al. (2015). Human-level control through deep reinforcement learning. *nature*, 518(7540):529–533.
- [Newsome et al., 1989] Newsome, W. T., Britten, K. H., and Movshon, J. A. (1989). Neuronal correlates of a perceptual decision. *Nature*, 341(6237):52–54.
- [Remijnse et al., 2006] Remijnse, P. L., Nielen, M. M., van Balkom, A. J., Cath, D. C., van Oppen, P., Uylings, H. B., and Veltman, D. J. (2006). Reduced orbitofrontal-striatal activity on a reversal learning task in obsessive-compulsive disorder. *Archives of general psychiatry*, 63(11):1225–1236.
- [Rolls, 2004] Rolls, E. T. (2004). The functions of the orbitofrontal cortex. *Brain and cognition*, 55(1):11–29.
- [Schultz et al., 1997] Schultz, W., Dayan, P., and Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306):1593–1599.

- [Squire and Zola-Morgan, 1991] Squire, L. R. and Zola-Morgan, S. (1991). The medial temporal lobe memory system. *Science*, 253(5026):1380–1386.
- [Sutton et al., 1998] Sutton, R. S., Barto, A. G., et al. (1998). *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge.
- [Tsividis et al., 2017] Tsividis, P. A., Pouncy, T., Xu, J. L., Tenenbaum, J. B., and Gershman, S. J. (2017). Human learning in atari. In *2017 AAAI spring symposium series*.
- [Ullman et al., 2012] Ullman, T. D., Goodman, N. D., and Tenenbaum, J. B. (2012). Theory learning as stochastic search in the language of thought. *Cognitive Development*, 27(4):455–480.
- [van Opheusden and Ma, 2019] van Opheusden, B. and Ma, W. J. (2019). Tasks for aligning human and machine planning. *Current Opinion in Behavioral Sciences*, 29:127–133.
- [Vertechi et al., 2020] Vertechi, P., Lottem, E., Sarra, D., Godinho, B., Treves, I., Quendera, T., Lohuis, M. N. O., and Mainen, Z. F. (2020). Inference-based decisions in a hidden state foraging task: differential contributions of prefrontal cortical areas. *Neuron*, 106(1):166–176.
- [Wickham, 2011] Wickham, H. (2011). The split-apply-combine strategy for data analysis. *Journal of statistical software*, 40:1–29.
- [Williams et al., 2004] Williams, Z. M., Bush, G., Rauch, S. L., Cosgrove, G. R., and Eskandar, E. N. (2004). Human anterior cingulate neurons and the integration of monetary reward with motor responses. *Nature neuroscience*, 7(12):1370–1375.

[Wilson et al., 2014] Wilson, R. C., Takahashi, Y. K., Schoenbaum, G., and Niv, Y. (2014). Orbitofrontal cortex as a cognitive map of task space. *Neuron*, 81(2):267–279.

[Zatorre et al., 2012] Zatorre, R. J., Fields, R. D., and Johansen-Berg, H. (2012). Plasticity in gray and white: neuroimaging changes in brain structure during learning. *Nature neuroscience*, 15(4):528–536.